WIRELESS TECHNOLOGIES

Introduction

There are several types of network which are used worldwide these days, both in houses and commercially. These networks are used on the bases of their scale and scope, historical reasons, preferences for networking industries, and their design and implementation issues.

LAN and WAN are mostly known and used widely.

LAN was first invented for communication between two computers. It operates through cables and network cards. Later WLAN, Wireless local area network was formed through LAN concept, there are no wires involved in communication between computers, and Wireless LAN cards are required to connect to wireless network.

LAN is the original network out of which other networks are formed according to requirements.

Classification of wireless network

LAN - Local Area Network

- LAN connects networking devices within a short spam of area, i.e. small offices, home, internet cafes etc.
- LAN uses TCP/IP network protocol for communication between computers. It is often but not always implemented as a single IP subnet.
- Since LAN is operated in a short area, it can be controlled and administrated by a single person or organization.

Wireless PAN (WPAN)

- It is a type of wireless network that interconnects devices within a relatively small area, generally within reach of a person as used by the likes of infrared and Blue-tooth technologies. For example, Blue-tooth provides a WPAN for interconnecting a headset to a laptop.
- Personal Area Networks (PAN) are formed by wireless communications between devices by way of technologies such as Blue-tooth and UWB.
- PAN standards are embodied by the IEEE 802.15 family of "Home Wireless" standards, which supersedes older infrared standards and HomeRF for dominance in this area of wireless communications.

Applications

PAN applications are virtually limitless. However, a key capability for a PAN is to enable devices to autonomously detect and acquire one another. This provides the ability for personalization through unconscious communications. For example, a PAN in ones automobile would provide various applications, such as the ability for the system to detect the presence of the user, thereby allowing the mobile handset to automatically acquire pertinent information for driving such as weather and road conditions.

One of the fundamental advantages of PANs is that they negate the need for wires,allowing the user to, for example, have a wireless headset, create ad hoc connectionsto other applications, include mobile commerce in which user of a mobile devicecommunicates with another machine for commerce such as ticket purchase, vendingand other small purchases.

The evolution of PANs will be from man-to-man, man-to-machine, and finallymachine-to-machine.

Issues and Challenges with PANs

The biggest initial issue will simply be to equip devices with software to enable the PAN connection. This will occur once technology such as Blue-tooth is costeffective and available in large quantities for deployment.

One of the biggest issues with PANs is the ability for devices to inter-operate withone another. This is not so much an issue with pre-established networks of devices,which all have the same vendor equipment, but it is a major issue for inter-vendor equipment connections. This is a major factor for unconscious communications.

Companies are working to solve these issues with various equipment mediationand interoperability software.

Ad hoc vs. Unconscious Communications

Ad hoc connections are simply those that occur in an automatic fashion, but the user is engaged in some activity in which it is known that a connection is about to occur. Examples include connecting to a LAN or buying a beverage from a vending machine.

On the other hand, unconscious communications occur as a result of automatic connections that are made between devices unknown to the user. Examples include a user coming home and the local base station in the home sensing this event and thus messaging to the network to treat all calls as home zone calls rather than mobilecalls in the macro network. Unconscious connections are the real end-goal for many applications that involve personalization and mobile commerce. The real issue will be to solve the interoperability problems between differing vendor provided equipment.

PAN Standards

_ 802.15.1a: Blue-tooth (2.4GHz at 1Mbps)

_ 802.15.2: Coexistence of PANs with one another

_ 802.15.3: High rate PAN (2.4GHz at 55 Mbps)

_ 802.15.3a: Alternative high rate PAN for UWB (2.4GHz at 110 Mbps)

_ 802.15.4: Low rate PAN - Zigbee

_ 802.15.4a: Alternative low rate - low power UWB

MAN - Metropolitan Area Network

- This kind of network is not mostly used but it has its own importance for some government bodies and organizations on larger scale.
- MAN falls in the middle of LAN and WAN.
- It covers a large span of physical area than LAN but smaller than WAN, such as a city .

CAN - Campus Area Network

- This kind of network is mostly used in relatively large universities or local business offices and buildings.
- Its networking is spanned with multiple LANs but smaller than a Metropolitan area network, MAN.

SAN - Storage Area Network

- SAN technology is used for data storage and it has no use for most of the organization but data-oriented organizations.

- Storage area network connects servers to datastorage devices by using Fiber channel technology.

SAN - System Area Network

- SAN, system area networks are also known as cluster area network and they connect high performance computers with high speed connections in cluster configuration.

Wireless LAN (WLAN)

- WLAN is a wireless alternative to a computer LAN that uses radio instead of wires to transmit data back and forth between computers in a small area such as a home, office, or school.
- WLANs are increasingly becoming an attractive alternative to licensed spectrum.
- Initially thought of strictly as a competitor to 3G, WLAN is now thought of as a complement to cellular based data services, providing a high bandwidth alternative to 3G at a fraction of the cost.
- Wireless LANs are standardized under the IEEE 802.11 series.

Wi-Fi:

- Wi-Fi is a commonly used wireless network in computer systems to enable connection to the internet or other devices that have Wi-Fi functionalities.
- Wi-Fi networks broadcast radio waves that can be picked up by Wi-Fi receivers attached to different computers or mobile phones.

Fixed Wireless Data:

- This implements point to point links between computersor networks at two locations, often using dedicated microwave or laser beams over line-of-sight paths. It is often used in cities to connect networks in two or more buildings without physically wiring the buildings together.

Being that WLAN's are only available through limited access points referred to as hot spots, it is important to recognize that WLAN service does not allow for truly ubiquitous service.

3G and WLAN

One of the more important issues associated with WLAN roaming is the ability to allow roaming between 3G and WLAN services. This is a key capability to allow for a seamless service experience between all the (licensed and unlicensed spectrumbased) services that a mobile network operator may offer.

Application Areas

- LAN Extension: Wireless LAN- linked into a wired LAN on same premises;Wired LAN Backbone, Support servers and stationary workstations; WirelessLAN-Stations in large open areas, Manufacturing plants, stock exchangetrading floors, and warehouses.
- Cross-building interconnect: Connect LANs in nearby buildings-a pointto point wireless link is used between two buildings. Device connected aretypically bridges and routers.
- Nomadic Access: Wireless link between LAN hub and mobile data terminalequipped with antenna; Laptop computer or notepad computer

Uses:

– Transfer data from portable computer to office server

– Can be useful in extended environment such as campus

– Ad hoc networking

Ad Hoc Networking - (no infrastructure): Temporary peer-to-peer networkset up to meet immediate needs; Example: Group of employees with laptopsconvenes for a meeting and later link computers in a temporary network forduration of meeting.

WAN - Wide Area Network

- WAN cover large distance for communication between computers. The Internet itself is the biggest example of WAN, which is covering the entire earth.
- WAN is a distributed collection of geographically LANs. A network connecting device router connects LANs to WANs.
- WAN uses network protocols like ATM, X.25, and Frame Relayfor long distance connectivity.

Wireless MAN

- Wireless Metropolitan Area Networks are a type of wireless network that connects several Wireless LANs.
- WiMAX (from "Worldwide Interoperability for Microwave Access") is the term which has been commercialized to refer to wireless MANs and is covered in IEEE 802.16d/802.16e.

PAN

- A personal area network (PAN) is a computer network used for communication among computer devices (including telephones and personal digital assistants) close to one's person. The devices may or may not belong to the person in question.
- Thereach of a PAN is typically a few meters.
- A wireless personal area network (WPAN) can also be made possible with networktechnologies such as IrDA, Blue-tooth, UWB, Z-Wave and ZigBee

WiMAX

WiMAX is new technology in mobile computing. It is yet to be deployed on huge scale worldwide unlike technologies like Wi-Fi.

WiMAX is short form of Worldwide Interoperability of Microwave Access. Its performance is similar to some extent like Wi-Fi but at higher speed, at great distance and for greater number of users.

WiMAX operates on 802.16 standards defined by IEEE, Institute of Electricand Electronic Engineering. IEEE 802.16 standard is known to be WMAN,Wireless Metropolitan Area Network.

Using WiMAX technology, broadband access can be provided to more rural and urban areas where wireless services cannot be provided by other sources like fiber optics or by other regular telephonic cables.

There are so many plus points of this technology out of which one is range, its range starts from minimum 30 miles,about 30 feet wireless connectivity is provided byWi-Fi technology. WiMAX range depends upon connectivity, if the connection is with Line of Sight (LoS) it can increase

from up to 75 miles. WiMAX offers a solution called “Last mile” it can be used as alternative of cable and DSL internet access.
WiMAX can provide data rate up to 70Mbps from larger distances which can be reached up 30 miles.
WiMAX operates on two frequencies.
1. IEEE 802 is a very flexible standard and provides standard addresses frequencies ranges from 10 GHz to 66 GHz.
2. IEEE 802.16 standard frequency states that it will also support low latency applications like internet, video, voice all together.

WiMAX technology is quite new, therefore few vendors are trying to develop infrastructureto develop the technology.
Vendors like Intel and Fujitsu are leading providers for WiMAX chipset, called SoC.
Customer premise equipment (CPE) aremanufactured by using these WiMAX chips SoCs, this enables the equipment toaccess WiMAX base station or Wireless Broadband Access (WBA).

WiMAX Features

- Long Range: Connectivity from long range is a much highlighted feature ofWiMAX.
- Mobility: Using its long-range feature, it provides connectivity on areas where other means of high-speed Internet are not available. The other major mobility feature of WiMAX technology is that connecting devices (Laptops, PDA etc) can move in certain area and still be connected to the network.
- Interfacing: WiMAX radio signals are broadcasted to its subscribers from base stations.
- Accessibility: Accessing WiMAX base station is not a difficult task; its long-range capability provides connectivity at home from base stations within therange. To get high speed internet, you only require to subscribe to WiMAXservice providers.

BLUE-TOOTH

Blue-tooth is a standard and communications protocol primarily designed for lowpower consumption, with a short range (1 meter, 10 meters, 100 meters) based on low-cost transceiver microchips in each device.

It is designed to operate in an environment of many users. Up to 8 devices can communicate in a small area network called piconet. Ten piconets can co-exist in the same coverage range ofthe Blue-tooth radio. For security, each link is encoded against eavesdropping andinterference.

Blue-tooth Application Areas

- Data and voice: Access points facilitate real-time voice and data transmissionsby providing effortless wireless connection of portable and stationarycommunication devices.
- Cable replacement: Blue-tooth eliminates need for numerous, often proprietarycable attachments connection for any kind of communication device.
- Ad hoc networking: Devices with Blue-tooth radio can establish connectionwith another when in range
- Other prevalent applications of Blue-tooth.
- Wireless control of and communication between a mobile phone and a hands-free headset or car kit. This was one of the earliest applications to becomepopular.
- Wireless networking between PCs in a confined space and where little bandwidthis required.
- Wireless communications with PC input and output devices, the most commonbeing the mouse, keyboard and printer.
- Transfer of files between devices with OBEX. (object exchange protocol)
- Transfer of contact details, calendar appointments, and reminders betweendevices with OBEX.

Blue-tooth vs. Wi-Fi in networking

- Blue-tooth andWi-Fi. Both use the unlicensed ISM (industrial, scientific andmedical) band and apply spread spectrum technology.

- Wi-Fi provides higher throughput and covers greater distances, but requiresmore expensive hardware and higher power consumption.
- They use the same frequency range, but employ different multiplexing schemes.
- While Blue-tooth is a cable replacement for a variety of applications, Wi-Fiis a cable replacement only for local area network access.

INFRARED LANS

Infrared LANS are strengths of waves over Microwave Radio. The Spectrum for infrared is virtually unlimited. Its equipment is inexpensive and simple. Infrared share some characteristics of visible light that makes it attractive for certain LANconfigurations. i.e. Infrared is diffusely reflected by light-colored objects thus it is possible to use ceiling reflection to achieve coverage for entire room coverage.

Infrared does not penetrate walls. It is more easily secured against eavesdropping and has less interference between different rooms

Drawbacks of Infrared Medium

- Indoor environments experience infrared background radiation from sunlightand indoor lighting
- Ambient radiation appears as noise in an infrared receiver, requiring transmittersof higher power.
- Increased transmitter power is limited by concerns of eye safety and excessivepower consumption
- LOS limitation (Limited range)